

On the Switching Parameter Variation of Metal-Oxide RRAM—Part I: Physical Modeling and Simulation Methodology

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Abstract—The variation of switching parameters is one of the major challenges to both the scaling and volume production of metal-oxide-based resistive random-access memories (RRAMs). In this two-part paper, the source of such parameter variation is analyzed by a physics-based simulator, which is equipped with the capability to simulate a large number (~ 1000) of cyclic SET-RESET operations. By comparing the simulation results with experimental data, it is found that the random current fluctuation experimentally observed in the RESET processes is caused by the competition between trap generation and recombination, whereas the variation of the high resistance states and the tail bits are directly correlated to the randomness of the trap dynamics. A combined strategy with a bilayer dielectric material and a write-verification technique is proposed to minimize the resistance variation. We describe the simulation methodology and discuss the dc results in Part I. The corroboration of the model and the device optimization strategy will be discussed in Part II.

Index Terms—Device modeling, dielectric breakdown, fluctuations, metal oxide, resistive switching memory (RRAM), switching parameter variation.

I. INTRODUCTION

THE METAL-OXIDE resistive random access memory (RRAM) is extensively studied as a competitive candidate for future nonvolatile memory application due to its simple structure, fast switching speed, great scalability, and compatibility with silicon complementary metal-oxide-semiconductor (CMOS) technology [1]–[3]. Among dozens of switching oxide materials, NiO [4], WO_x [5], HfO_x [6], [7], TaO_x [8], [9], and AlO_x [10]-based memory devices have shown excellent reliability, including long endurance and stable retention at elevated temperature. A 4-Mb macrochip based on HfO_x mem-

ory has been demonstrated [11]. However, the key obstacle for metal-oxide RRAM to becoming mature for large-scale manufacturing is relatively poor control of switching uniformity. Significant parameter fluctuations exist in the switching voltages, as well as the resistance distributions, which include temporal fluctuations (cycle to cycle) and spatial fluctuations (device to device). A large memory array usually contains noticeable tail bits in the resistance distribution [12], which remarkably reduces the resistance window and therefore imposes a great challenge to realizing the multilevel capability of the metal-oxide RRAM technology.

The origin of the variations in the resistive switching is not well understood yet, despite the fact that the variations are observed in all RRAMs reported. Phenomenological models such as the network of random circuit breakers [13] have been applied to unipolar¹ RRAM devices. In this paper, this problem is studied for the metal-oxide-based bipolar RRAM devices. Through the development of a physics-based stochastic simulator to quantify the switching variations, we identify the major sources of variations and fluctuations. The obtained insights provide guidelines for future device design and optimization against the undesired switching parameter variations. In Part I of this paper, we describe the developed model, which includes the trap-assisted-tunneling (TAT) current solver and the stochastic generation and recombination of oxygen vacancies (Vo). The considerations and approximations to reproduce the switching characteristics of a bipolar RRAM device are discussed. In Part II of the paper [14], we corroborate the simulation results with the experimental data of HfO_x -based RRAM devices [15], [16], such as dc I - V switching characteristics and pulse transient switching waveform, multilevel resistance distribution, etc., with a focus on discussing the origin of the tail bits problem and possible elimination methods such as the use of the write-verify technique and a bilayer oxide device structure.

II. MODEL DESCRIPTION

A widely recognized bipolar switching physical picture [17] is shown in Fig. 1. The transition between the high resistance state (HRS) and the low resistance state (LRS) has been

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¹Here, unipolar means the set from the HRS to the LRS and the reset from LRS to HRS occur at the same voltage polarity, and bipolar means the set and the reset occur at opposite voltage.

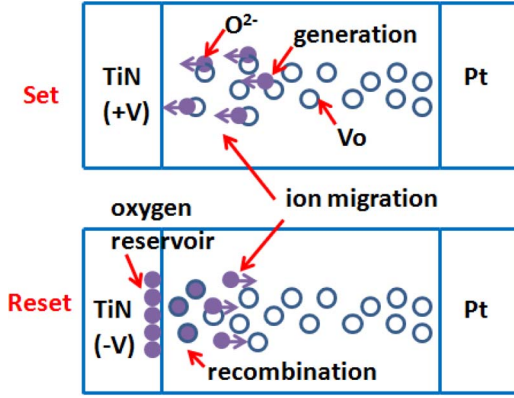


Fig. 1. Schematics of the SET/RESET mechanisms in a metal-oxide-based RRAM cell. In a SET process, oxygen ions are pulled out of the lattice and drift to the active electrode, leaving behind the conducting filaments formed by Vo, which causes the decrease in resistance. When a reverse bias is applied in a RESET process, the stored oxygen ions at the electrode/dielectric interface migrate into the dielectric, recombining with the Vo and rupturing the filaments, leading to an increase in resistance.

attributed to the formation and rupture of conductive filaments (CFs) in the oxide, which may consist of Vo (serving as traps) or metal precipitates [18], [19]. Joule heating [20] and electric-field-assisted migration of oxygen ions/vacancies [21] have been suggested to play an important role in the switching. Specifically, the FORMING/SET process of the memory cell (transition from HRS to LRS) is interpreted as a dielectric soft breakdown associated with the generation of oxygen vacancies and the migration of oxygen ions toward the electrode/oxide interface (where they are stored), leaving behind the oxygen vacancies or metal precipitates in the bulk oxide to form percolation-conducting paths [20], [21]. The RESET of the device is associated with the process in which the oxygen ions from the anode/dielectric interface migrate back into the dielectric layer, recombining with the oxygen vacancies (or equivalently, oxidizing the metal precipitates) and thus rupturing the conducting filaments [20], [21].

In essence, the generation/recombination of oxygen vacancies is a stochastic process that is similar to the well-known randomness of dielectric breakdown [22]. The description of the resistive switching of oxide-based RRAM must take into consideration the behaviors of both electrons and ions and reflect the nature of this randomness. To restrict the simulation within a reasonable computation resource, the adiabatic approximation is adopted, which allows us to describe the behaviors of electrons and ions in a decoupled manner by treating the conductance of the cell as a function of Vo positions (which will be referred to as *Vo configurations* hereafter). The electronic part involved in the operation of the cell has been discussed in an experimental work in [23] and [24], which concludes that the TAT through the Vo is the dominant conduction mechanism in the cell. Similar mechanisms have also been identified by the first principle study of RRAM cells [25]. Based on these studies, a TAT current solver is developed in this work to evaluate the current and the resistance at a given Vo configuration. The Vo configurations determined by ion dynamics are simulated based on the migration velocity of ions and the generation/recombination probability of Vo.

A programming cycle (i.e., FORMING/SET/RESET) of the memory cell is divided into multiple time steps, each assumed to be a quasi-steady process. At the end of each time step, the Vo configuration is updated according to the Vo dynamics previously described, which is used by the TAT solver to calculate the electrical current. The temperature of the cell is then updated for the calculation of configuration in the next step.

We now discuss the TAT solver and Vo dynamics in more detail.²

A. Electron Transport by the TAT-Solver

In the dielectric region of a HfOx-based RRAM, the current continuity equation is in the form of [26]

$$\frac{df_n}{dt} = (1 - f_n) \sum_{m=1}^N R_{mn} f_m - f_n \sum_{m=1}^N R_{nm} (1 - f_m) + (R_n^{iL} + R_n^{iR}) (1 - f_n) - (R_n^{oL} + R_n^{oR}) f_n \quad (1)$$

with $f_n \in [0, 1]$ being the electron occupation probability of the n th trap, R_{mn} being the electron hopping rate from trap m to n , R_n^{oL}/R_n^{oR} being the electron hopping rate from trap n to the left/right electrode, and R_n^{iL}/R_n^{iR} being the electron hopping rate from the left/right electrode to trap n . Σ' stands for the summation over all integers, except n . As reported by different authors, the oxygen vacancies, which are positively charged to +1, are thermodynamically favored in TAT [24], [27], [28].

In a quasi-steady state, (1) becomes

$$(1 - f_n) \sum_{m=1}^N R_{mn} f_m - f_n \sum_{m=1}^N R_{nm} (1 - f_m) + (R_n^{iL} + R_n^{iR}) (1 - f_n) - (R_n^{oL} + R_n^{oR}) f_n = 0. \quad (2)$$

The current flowing through the cell can be calculated by evaluating the electron flux through a cross section near, for example, the left electrode, i.e.,

$$I^L = -q \sum_{n=1}^N [(1 - f_n) R_n^{iL} - R_n^{oL} f_n]. \quad (3)$$

Summing (2) over n , it can be proved that $I = I^R = I^L$.

The hopping rate between two vacancies can be calculated by the Mott hopping model [29], [30]

$$R_{nm} = R_0 \exp \left[-\frac{r_{nm}}{a_0} + \frac{qV_{mn}^H}{kT} \right] \quad (4)$$

with $R_0 \approx 10^{12}$ Hz being the vibration frequency of an electron, $r_{nm} = |r_m - r_n|$ being the distance between the two vacancies, a_0 being the attenuation length of the electron wave function, $V_{mn}^H \approx -F^H(r_m - r_n) \approx V^H(r_m) - V^H(r_n)$

²In the subsequent discussions of this paper, Vo and traps are interchangeable concepts both representing the oxygen vacancies.

TABLE I
PARAMETERS USED IN THIS WORK TO SIMULATE THE HfO_x-BASED RRAM

Symbol	Quantity	Value	Equation No.
R_0	Vibration frequency of an electron in a trap	10^{12} Hz	(4)
a_0	Attenuation length of the electron wave function in a trap	0.33 nm	(4)
$R_{\text{tunnel}}^0 N^{L,R}$	Electronic coupling factor between the electrode and the dielectric layer	10^{14} Hz	(5)
m^*	Tunneling effective mass	$0.1m_0$	(7)
$E_C - E_{v0}^+$	Trap energy of an empty Vo below the conduction band of HfOx	1.83eV	(7)
$E_C - E_{v0}^\bullet$	Trap energy of a filled Vo below the conduction band of HfOx	1.97eV	(7)
$E_F^{L,R}$	Fermi level of the left/right electrode with respect to the conduction band of HfOx	-1.9eV	(8), (9)
$1/t_0$	Characteristic vibration frequency of oxygen ions in HfOx	10^{13} Hz	(10)
E_a	Average migration barrier height of oxygen ions in HfOx	1 eV	(10)
γ	Enhancement factor of the electric field in lowering the migration barrier	3 (FORMING, SET), 0.3 (RESET)	(10)
β_0	Recombination enhancement factor due to the presence of excessive oxygen ions	3×10^3	(13)
L_p	Concentration decaying length of oxygen ions	1 nm	(13)
γ^{drift}	Drift coefficient of oxygen ions due to an external field	8 (DC-sweep), 11 (pulse)	(14)
E_m	Migration barrier of oxygen ions	1.0 eV	(14)
R_{th}	Equivalent thermal resistance of HfOx	5×10^5 K/W	(21)

being the change of barrier height due to the applied external field [31], and $V^H(r_n)$ being the homogeneous component of the local electric potential to be described in Section II-C.

The hopping rates between a trap and an electrode are

$$\begin{aligned}
 R_n^{\text{il},R} &= R_{\text{tunnel}}^0 N^{L,R} (E_v^+) F_{\text{in}}^{L,R} (E_v^+) T_n^{L,R,+} \\
 R_n^{\text{OL},R} &= R_{\text{tunnel}}^0 N^{L,R} (E_v^\bullet) F_{\text{out}}^{L,R} (E_v) T_n^{L,R,\bullet}
 \end{aligned} \quad (5)$$

where R_{tunnel}^0 is the coupling strength between the trap and the electrode, with the unit of a tunneling rate. $N^{L,R}$ is the number of states at the given energy in the electrode. The product of $R_{\text{tunnel}}^0 N^{L,R}$ is fitted to experiments for a specific con-

tact electrode and dielectric material (value listed in Table I). E_v^+ ($E_v^\bullet$) is the energy of an empty (a filled) trap determined by

$$E_v^{+,\bullet}(r_n) = E_{v0}^{+,\bullet} - qV^H(r_n) \quad (6)$$

with E_{v0}^+ ($E_{v0}^\bullet$) located below the conduction band of HfO_x. $T_n^{L,R,+,\bullet}$ is the tunneling probability from the left/right electrode into the trap, given by the Wentzel–Kramers–Brillouin approximation in (7), shown at the bottom of the next page, with x_n being the x th component of r_n , L being the thickness of the dielectric layer, and $m^* = 0.1m_0$ being the tunneling effective mass for HfO_x [32]. $F_{\text{in}}^{L,R}(E_v^+)$ is the Fermi integral

representing the number of filled states in the electrode above E_v^+ , which can inject electrons into trap n , i.e.,

$$\begin{aligned} F_{\text{in}}^{L,R}(E_v^+) &= \int_{E_{v0}^+ - qV(x_n)}^{+\infty} f[E - (E_F^{\text{LR}} - qV^{\text{LR}})] dE \\ &= \int_{E_{v0}^+ - qV(x_n)}^{+\infty} \frac{1}{1 + \exp[(E - (E_F^{\text{LR}} - qV^{\text{LR}}))/kT]} dE \end{aligned} \quad (8)$$

with E_F^{LR} being the Fermi level in the left/right electrode, and V^{LR} being the applied bias on the left/right contact. Similarly, $F_{\text{out}}^{L,R}(E_v^-)$ is the Fermi integral representing the number of empty states in the electrode below E_v^- , which can accept electrons from trap n , i.e.,

$$\begin{aligned} F_{\text{out}}^{L,R}(E_v^-) &= \int_{-\infty}^{E_{v0}^- - qV(x_n)} \{1 - f[E - (E_F^{\text{LR}} - qV^{\text{LR}})]\} dE \\ &= \int_{-\infty}^{E_{v0}^- - qV(x_n)} \frac{1}{1 + \exp[(E - (E_F^{\text{LR}} - qV^{\text{LR}}))/kT]} dE. \end{aligned} \quad (9)$$

Note that, due to the limit of computation resource, we have assumed that, with the assistance of phonons, all the electrons in the electrode above E_v^+ can be injected into trap n with the same probability as an elastic tunneling process. Similarly, all the empty states in the electrode below E_v^- are assumed to be able to accept injected electrons from trap n with a constant probability. A more rigorous and computationally demanding description of inelastic tunneling will inevitably require the consideration of electron-phonon interactions for all the hopping events, thus introducing another factor in (4) and (5), respectively (see, e.g., [30] and [33]). The simplification in our simulation by lumping the inelastic factor into the fitting parameters of R_0 and R_{tunnel}^0 may not fully account for the temperature dependence of phonon-assisted tunneling. It is adopted here mainly to improve computational efficiency.

For a given Vo configuration, the hopping rates are evaluated by (4)–(9) and substituted into (2), which then can be solved using a Newton–Raphson iteration scheme.

B. Generation/Recombination of Oxygen Vacancies

When an external field is applied to the dielectric material, the oxygen ions can be pulled out of their equilibrium position, generating the oxygen vacancies. The generation probability of an oxygen vacancy on each lattice site (mesh point) during a time interval $[\tau, \tau + t]$ is approximated as

$$P_G(F^{\text{eq}}, T, t) = \frac{t}{t_0} \exp[-(E_a - \gamma|F^{\text{eq}}|)/kT] \quad (10)$$

where $1/t_0 = 10^{13}$ Hz is the characteristic vibration frequency of the oxygen ions [34]. $E_a \approx 1$ eV is fitted as the height of the migration barrier, which an oxygen ion must overcome to break out from its equilibrium position, which is close to the reported values from experiments [35], [36]. F^{eq} is the local electric field felt by the ion in its equilibrium position, and γ is a coefficient representing the local enhancement factor due to the electric field [37]. It should be noted that this generation process depends only on the magnitude of the electric field and may happen during the entire operation cycle of a memory cell, including the RESET process when oxygen vacancies are being annihilated.

The reverse process of Vo generation is Vo recombination. This mainly happens in a RESET process, during which the oxygen ions are pulled back from the electrode/dielectric interface and captured by the Vo inside the dielectric layer. In equilibrium state with no field applied, the recombination events balance with the generation events; thus

$$P_R^0 = P_G(F^{\text{eq}} = 0, T, t) = \frac{t}{t_0} \exp(-E_a/kT). \quad (11)$$

When excess oxygen ions are present, Vo recombination is given by

$$P_R = \beta P_R^0 \quad (12)$$

where $\beta \propto C_{o2-}$, with C_{o2-} being the concentration of oxygen ions. Ideally, the concentration of oxygen ions should be obtained from the numerical solution of the ion transport equation [21]. However, this becomes computationally expensive when a large number of operation cycles need to be simulated. To achieve a balance between accuracy and efficiency, we approximate β with the following analytical form:

$$\beta(x, t) \approx \beta_0 \exp(-vt/L_p) u(x, t) \quad (13)$$

where L_p is the decaying length of ion concentration, and v is the field-driven wave-front velocity of oxygen ions determined

$$\begin{aligned} T_n^{L+, \bullet} &= \exp \left\{ -2 \int_0^{x_n} \frac{1}{\hbar} \sqrt{2m^* [E_C - E_{v0}^{+, \bullet} - qV^H(x) + qV^H(x_n)]} dx \right\}, & E_v^{\pm}(x_n) < E_C - qV^H(x) \\ T_n^{R+, \bullet} &= \exp \left\{ -2 \int_{x_n}^L \frac{1}{\hbar} \sqrt{2m^* [E_C - E_{v0}^{+, \bullet} - qV^H(x) + qV^H(x_n)]} dx \right\}, & E_v^{\pm}(x_n) < E_C - qV^H(x) \end{aligned} \quad (7)$$

by [21]

$$v = \frac{a}{t_0} \exp(-E_m/kT) \sinh(q\gamma^{\text{drift}} F/kT) \quad (14)$$

with $F = -\nabla V^H$ being the electric field felt by a mobile oxygen ion, E_m being the migration barrier for oxygen ions, $a \approx 0.25$ nm being the lattice constant (also the mesh size), and γ^{drift} being the enhancement coefficient related to the local dielectric constant of the material [37]. $u(x, t)$ describes the diffusive behavior of oxygen ions and can be approximated by a complementary error function. In this paper, it is tabulated as

$$u(x, t) = \begin{cases} 1, & x \leq vt \\ 0.3, & vt < x \leq vt + a \\ 0.1, & vt + a < x \leq vt + 3a \\ 0, & x > vt + 3a. \end{cases} \quad (15)$$

At the end of each simulation time step, (10) and (12) are evaluated to determine the Vo configurations. Then, the current is evaluated by the TAT solver. We have used different values of the enhancement factor for ion drift (γ^{drift}) and vacancy generation (γ) and for different operation processes (SET/RESET). It is found that, although the theoretical value of the enhancement factor can be related to the permittivity of the material [37], using a single value can hardly make the simulation result agree with measured I - V . We believe that this is due to the variation of local permittivity at different resistance states and different pulse frequencies. Therefore, the enhancement factor is separately fitted for different processes (values summarized in Table I).

C. Electrical Potential and Temperature

The electrical potential inside the cell satisfies the Poisson equation

$$-\nabla^2 V = \frac{\rho}{\varepsilon} \quad (16)$$

with the boundary conditions defined by the applied voltage at the contact electrodes

$$V(x=0) = V^L \quad V(x=L) = V^R. \quad (17)$$

The solution can be divided into the homogeneous component V^H and the nonhomogeneous component V^N . V^H satisfies the homogeneous counterpart of (16) and the nonhomogeneous boundary condition of (17), i.e.,

$$\begin{cases} -\nabla^2 V^H = 0 \\ V^H(x=0) = V^L \\ V^H(x=L) = V^R \end{cases} \quad (18)$$

whereas V^N satisfies the nonhomogeneous equation of (16) and the homogeneous counterpart of (17), i.e.,

$$\begin{cases} -\nabla^2 V^N = \frac{\rho}{\varepsilon} \\ V^N(x=0) = 0 \\ V^N(x=L) = 0. \end{cases} \quad (19)$$

Therefore, V^N can be understood as the local Coulomb potential due to the presence of partially depleted Vo, and V^H can be

understood as the change of local potential superimposed onto V^N due to the applied external bias at the boundaries. According to [31], the change of the local barrier height between two traps is proportional to the homogeneous component (i.e., V^H) to the first order. Therefore, the solution of (18) is used in each time step to compute the trap-to-trap hopping probability in (4).

The electric field F^{eq} in (10) is different from the change of barrier height previously discussed, in that it is the field felt by an oxygen ion in its equilibrium position. In this paper, we have adopted a simplified picture and assume the following: 1) The field felt by the ions are along the direction normal to the electrode pads. 2) The formed filaments can be approximated as bodies with equipotential. Assuming that the F^{eq} is along the x -direction, this enables us to estimate the force on an oxygen ion by

$$|F^{\text{eq}}(x=ia, y=ja)| = \frac{|V^R - V^L|}{L - a \sum_{i=1}^M \delta_{ij, \text{Vo}}} \quad (20)$$

where i and j are mesh indices. $\delta_{ij, \text{Vo}}$ takes the value of 1 if there is a Vo at site (i, j) or the value of 0 otherwise. This approximation has neglected the nonnormal field component around the filament tips. However, considering the fact that the underestimated field components are relatively small as compared to the local variation of material parameters and field and that the growth of filaments along a nonorthogonal direction do not have a first-order contribution to the cell's resistance, neglecting these additional components is not expected to lead to a substantially different behavior of the statistical distribution of the final resistance.

The local temperature plays an important role in both SET and RESET processes. Generally, the temperature distribution can be obtained by solving the heat transfer equation, with multiple heat sources due to the inelastic electron-phonon interactions [38]. In this paper, we have adopted a simplified picture, in which the temperature is assumed to be uniform in the vicinity of the filaments and is correlated to the Joule heating by [20]

$$T = 297 \text{ K} + |(V^L - V^R)I| R_{\text{th}} \quad (21)$$

where V^L and V^R are the applied bias on the two electrodes, I is the current flowing through the nearby filaments, and R_{th} is an equivalent thermal resistance representing the impact of Joule heating on the temperature [20]. This leads to a typical increase of temperature during the SET process by around 170 K.

All the parameters used in (4)–(21) for our simulation are listed in Table I.

D. Implementation of a Current Compliance

A current compliance is normally required during the SET and FORMING processes to control the uniformity of the resistance during different cycles. The current compliance can be enforced, e.g., by a semiconductor parameter analyzer or a selection transistor in series with the RRAM. There are multiple options to implement this compliance in a simulation, depending on the test configuration in the experiments. In this

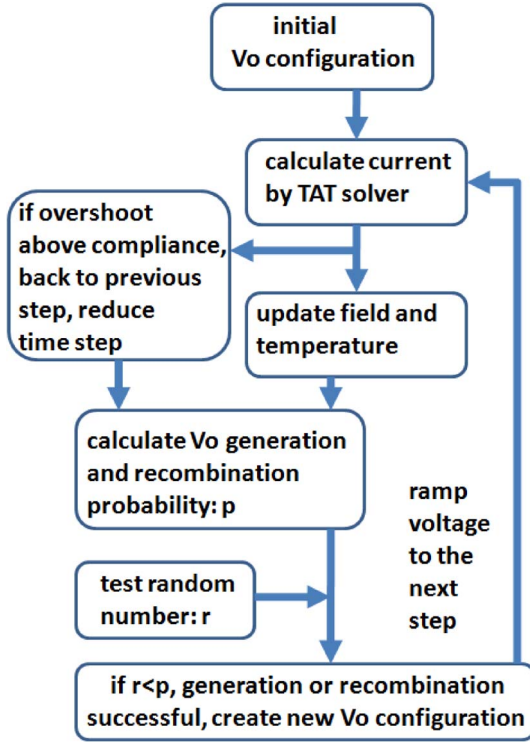


Fig. 2. Flow chart of the fast simulator developed in this work for the study of switching parameter variations.

paper, we have adopted a trial-and-error strategy to ensure that the current overshoot is within a specified tolerance above the current compliance. At the beginning of a new time step, the latest Vo configuration is stored. A trial time step is used to generate the Vo configuration and current. If the current is below the compliance plus the tolerance, the step size is accepted. Otherwise, the initial guess is rejected, and a smaller step size is used to generate the Vo configuration and current again. The test loop goes on until the step size is sufficiently small, so that the current is below the compliance plus the tolerance. The final accepted time step corresponds to the characteristic response time of an external current limiter or the transient time determined by the parasitic capacitance. Once the current exceeds the compliance, the SET or FORMING process will stop after the next accepted time step.

E. Simulation Flow

Equations (1)–(21) apply to 1-D–3-D systems. The choice of dimension is a function of the simulation purpose. Because the goal of this work is to study the variation and fluctuation of device parameters, which requires a large number of repeated simulations, we have implemented the model in a 2-D memory cell, which includes sufficient stochastic property of an RRAM cell while keeping the simulation within acceptable computation load. The physics along the third dimension is lumped into the number of states ($N^{L,R}$) in (5), which is fitted to experimental results to minimize the impact due to the lack of freedom along the third dimension. The simulation flow is shown in Fig. 2, which applies to the processes of FORMING, SET, and RESET, and can be configured to be either a dc-sweep

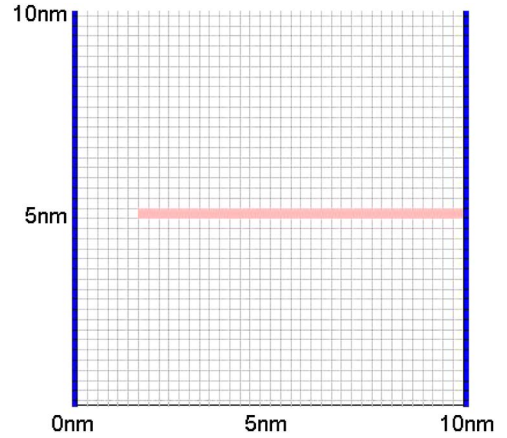


Fig. 3. One-dimensional single vacancy chain formed by oxygen vacancies. The vacancy chain is connected to the right electrode but disconnected to the left electrode with a gap in between.

analysis (in which the input voltage increases or decreases linearly with time) or a pulse programming cycle for statistical analysis (in which the input voltage is a rectangular pulse).

In the FORMING process, an initial Vo configuration is generated by randomly placing a specified number of Vo in the simulated region, representing the weak spot of an as-fabricated HfO_x RRAM cell. The voltage across the cell then ramps up with a rate of 1 V/s in a dc sweep. In a pulse operation, a pulse voltage is applied instead with a limited time width. The TAT solver and the Vo dynamics are repeatedly invoked to check the resistance of the cell during the process, until the specified compliance is reached with the trial-and-error strategy previously explained.

The RESET process begins with a Vo configuration after the FORMING or the previous SET process, which corresponds to an HRS. A reverse bias is applied to the cell, and the voltage ramps up with a rate of 1 V/s. (In a pulse operation, a reverse pulse voltage is applied instead with a limited time width.) The wavefront of oxygen ions begins to move into the cell when the temperature and field are sufficiently high. The recombination dominates the generation due to the high oxygen concentration in the ruptured region, causing the current to drop when the percolation paths or filaments are ruptured. The process ends when a specified voltage maximum is reached and the dc voltage sweeps back to zero, or when the applied pulse has ended.

The SET process is similar to the FORMING process, except that it starts from the Vo configuration after the previous RESET process, where the filaments in the cell are partially ruptured.

The SET/RESET processes are concatenated and repeated by multiple times in the simulation to study the distribution of HRS and RESET voltage.

III. ONE-DIMENSIONAL VO CHAIN

Before discussing the dc I - V characteristics and variations in a 2-D HfO_x RRAM cell, it is inspiring to discuss the model's observations for a 1-D Vo chain first. Consider the case in which there is only one chain of Vo between the two electrodes of an RRAM cell, as shown in Fig. 3. The conducting current has been mainly attributed to the electrons that tunnel from

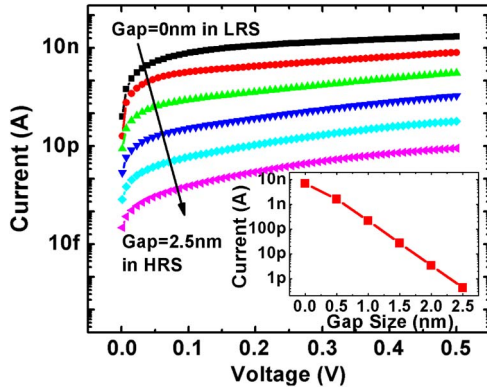


Fig. 4. I - V characteristics of the 1-D vacancy chain with different gap sizes. The inset shows the exponential decrease in current with increasing gap distance, which explains the HRS variation.

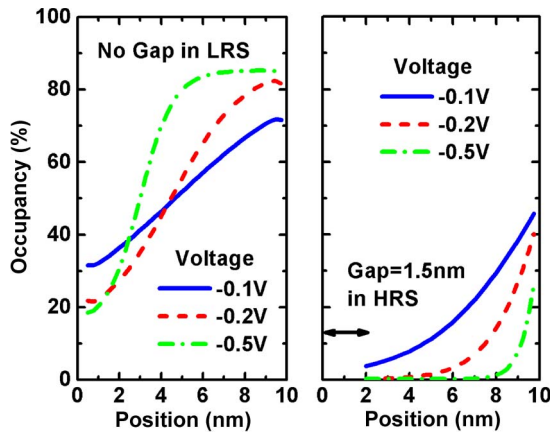


Fig. 5. Electron occupancy along a 1-D vacancy chain under different bias voltages in LRS and HRS. In the HRS, the increase in voltage causes depletion to the traps near the cathode and makes the cathode-to-trap tunneling the bottleneck of the entire conduction path.

the cathode into the closest trap in the dielectric, then hop through the chain of traps, and finally tunnel into the anode. Consider the case in which a test voltage is applied to the cell such that the left and right electrodes serve as the cathode and anode, respectively. In addition, assume that the chain is connected with the anode, whereas it is disconnected from the cathode with a certain gap. The electron from the cathode must tunnel through the ruptured gap before it hops through the chain to the anode on the right side. Conduction through this chain is therefore determined by the bottleneck in the ruptured region. Since the tunneling probability exponentially depends on the gap size, the current through the cell has an exponential dependence on the ruptured gap size, which is verified by the output of the TAT solver in Fig. 4. The variation in the gap size therefore is expected to exert a significant impact on the HRS.

Fig. 5 shows the electron distributions in the traps for both a complete V_o chain connecting both electrodes and a ruptured V_o chain previously described. It is observed that an increasingly negative (reverse) bias on the left electrode (which works as a cathode) drives the electrons to the right electrode (which works as an anode). In the case of a complete V_o chain, this causes the electrons to pile up near the anode region. In the

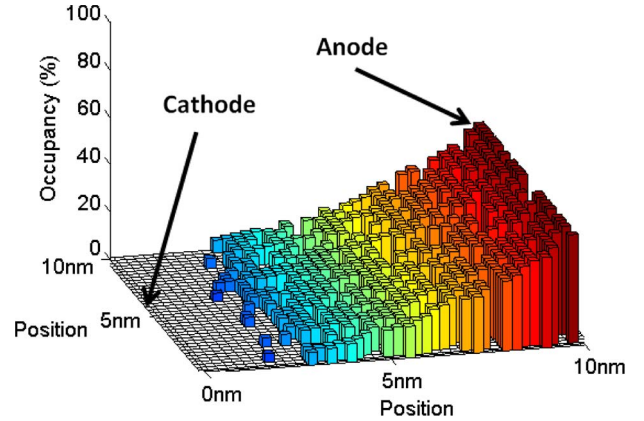


Fig. 6. Two-dimensional electron occupancy in the HRS when a reverse bias is applied. The front line of filaments is not straight due to the stochastic nature of V_o generation and recombination. The variation of the gap distance is the origin of the HRS variation.

other case where the chain is partially ruptured near the left electrode, the gap between the left electrode and the leftmost trap becomes a bottleneck for conduction. Therefore, under a reverse bias, the traps near the left electrode (cathode) tend to be *depleted* in order to accept sufficient electron flux from the cathode to retain the current continuity [see (1)]. The depleted V_o are more positively charged and hence more likely to capture the negatively charged oxygen ions. Therefore, the V_o near the cathode has higher probability to be recombined during a RESET process.

IV. DC SWEEP ANALYSIS

Fig. 6 shows the electron occupancy in a 2-D case, where the gap size varies along the lateral direction of the cell. In the following sections of this paper, we discuss the application of the model for a 2-D HfO_x RRAM cell. More results with experimental corroboration on RESET current fluctuation, RESET voltage variation, and the HRS tail bits will follow in Part II of this paper.

Fig. 7(a) shows an as-fabricated metal- HfO_x -metal structure simulated in this work. The thickness of the dielectric layer is 10 nm, with two electrodes on the left and right boundaries. The simulated region is 10 nm along the pad direction and 10 nm along the thickness direction, which can be regarded as a weak portion of an experimental memory cell, e.g., the grain boundary or the as-fabricated defect rich region, where the filaments concentrate [24]. We will discuss the FORMING, RESET, and SET operations in a sequential order according to the dc simulation flow.

A. Forming

The as-fabricated memory cell is in an HRS with no established filaments but multiple randomly distributed V_o as defects. During the FORMING process, a positive voltage is applied on the active electrode (in this case, the left electrode), building up a strong field, which increases the probability of V_o generation. The filaments start to grow around the initial V_o when the voltage is sufficiently high. Formed segments of

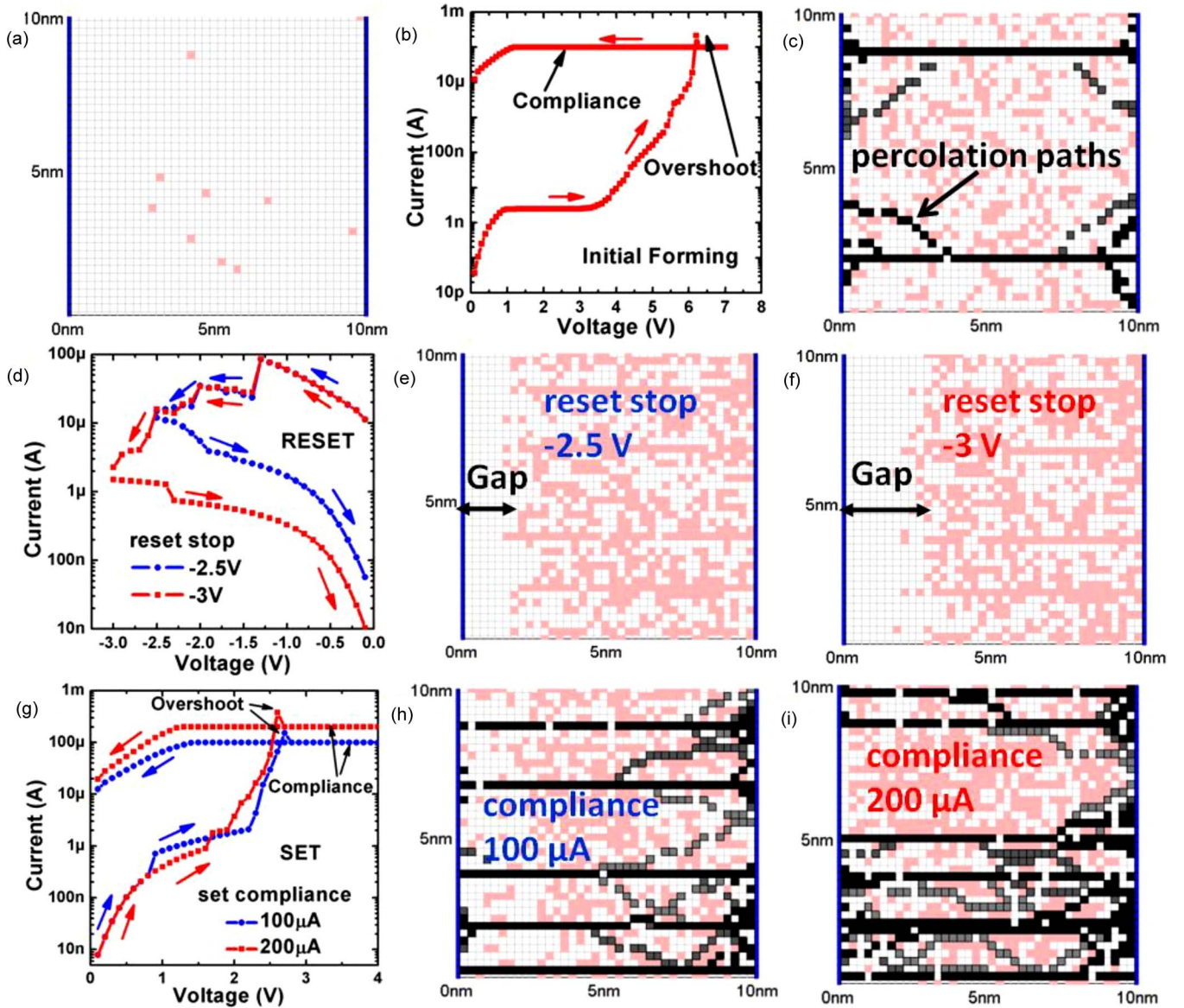


Fig. 7. DC sweep simulation of a $10 \text{ nm} \times 10 \text{ nm}$ RRAM region, corresponding to the weak spot of the cell. (a) As-fabricated state of the region, with the V_o randomly distributed in an initial configuration. (b) I - V characteristics of a FORMING process. (c) Percolation paths after forming. (d) I - V characteristics of RESET with different stop voltages. (e) and (f) Smaller/larger gap sizes due to the smaller/larger RESET stop voltages. (g) SET processes with different current compliances. (h) and (i) Fewer/more V_o and percolation paths due to the smaller/larger current compliance. Percolation paths are identified by the Dijkstra shortest path algorithm and color coded according to the conducting strength (darker means stronger).

the filament are approximated by equipotential bodies, making the field in the gap regions between the segments stronger [see (20)]. In addition, the increased current causes the temperature to rise by the Joule heating effect in (21). Both effects lead to the faster generation of V_o , which causes further increase in field and temperature. The positive feedback shows up in the I - V curve in Fig. 7(b) as a steep slope, which finally exceeds the current compliance and causes a current overshoot. After FORMING, the simulated weak spot of the cell contains more V_o than the initial state. Applying the Dijkstra's shortest-path algorithm [39] allows us to visually identify the most conducting percolating paths (which form a filament) in the layer. We would like to point out that the formalism used in this work does not fully include the correlation between different vacancy generation events due to the lack of knowledge of such

correlation strength. Therefore, more work is required if the details of correlated growth of filaments should be resolved.

B. Reset

In a RESET process, a negative voltage is applied on the left electrode, making it a cathode from which the accumulated oxygen ions start to drift into the dielectric layer. When the ions get close to the depleted V_o , they recombine with V_o , resulting in the filament dissolution and an increase in resistance of the simulated structure. Fig. 7(d) shows the I - V characteristics of two RESET processes, which start from the same V_o configuration but sweeps to two different voltages. Key observations include the following: 1) A larger RESET voltage results in a larger gap distance and a correspondingly higher resistance

after the RESET. 2) A fluctuation characterized by multiple rising and falling steps of the current exists during the gradual resistive switching. 3) The resulting V_0 configurations shown in Fig. 7(e) and (f) feature a randomness in the front line of the ruptured filaments.

The first observation can be understood by noticing the fact that the length of the rupture region of the filaments is a function of voltage and time. With a constant ramp-up rate, sweeping to a higher stop voltage in a RESET process leads to both higher velocity and longer drift time of ions. In addition, the difference in velocity may be further amplified by the difference in the temperature. Therefore, a higher stop voltage in a dc sweep generally results in a larger average gap, leading to a higher resistance after RESET.

The second observation of current fluctuation is related to the competition between V_0 generation and recombination. Although V_0 recombination dominates in a RESET process, the high field may also induce V_0 generation in the gap region. This is particularly prominent in the beginning phase of the current drop when the gap size is small and the field and temperature are both high. The generated V_0 in the gap reconnect the ruptured filament with the cathode, giving rise to a momentary increase in current during RESET. The newly generated V_0 are mostly recombined soon, because the recombination probability is much higher than that of generation due to the overwhelmingly high concentration of oxygen ions present in the rupture gap. Therefore, the current drops again, resulting in a random fluctuation superimposed on the decreasing I - V curves. A similar fluctuation of current is also observed in a pulse RESET operation in both simulation and experiments. The results and corresponding impacts on the switching parameter variation will be shown and discussed in Part II of this paper.

The third observation on the random edge of the gap region is attributed to the stochastic nature of V_0 recombination. In a large number of RESET tests, the average gap distance is expected to follow a normal distribution, which leads to a normal distribution of the HRS in the log scale according to the exponential dependence of resistance on the gap distance, as discussed in the simplified 1-D V_0 chain model. Understanding this correlation between the variation of HRS and the randomness of the gap distance is the key to the optimization of the cell structure. Detailed design options will be discussed in Part II of this paper.

C. Set

The SET process is simulated in a similar way as the FORMING process, except that it starts from the V_0 configuration after the previous RESET. Fig. 7(g) shows the I - V characteristics of two SET processes, which start from the same V_0 configuration but have two different current compliance levels. It is observed that a larger compliance level generally results in more V_0 (thus more filaments) after the SET process. This agrees with a recent experimental measurement [40]. Therefore, the restriction of current overshoot is important to the control of variation in the LRS resistance.

Comparing the V_0 configurations after SET and FORMING, it is observed that the V_0 distribution after FORMING is more

random than those after several SET and RESET cycles. This implies that filaments tend to regrow from the same weak spots in different SET cycles, which explains the stabilization of the SET voltage after the first few cycles. Experiments have also shown that the initial FORMING process generally requires a high voltage than the stabilized SET voltage. From the simulation, it is clear that this is because FORMING starts from a configuration with fewer V_0 , whereas SET starts from a configuration with partially ruptured filaments.

V. CONCLUSION

A hybrid methodology has been developed in this work to simulate the resistive switching of metal-oxide RRAM cells. The simulation framework includes both a deterministic TAT solver for the electron transport and a stochastic description of the V_0 dynamics during the switching processes. Combining both aspects enables us to simulate the complete dc and pulse switching behaviors of the RRAM cell. Despite the simplification along the third dimension and the assumptions of uniform material parameters and the global temperature, the approach successfully reveals the correlation between the terminal I - V curves and the stochastic V_0 dynamics inside the switching material. The development of the simulation tool opens up the possibility to systematically study the variation of switching parameters in the metal-oxide RRAM cells, as will be discussed in Part II of this paper.

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